

Car Plate recognition and reconstruction with Deep Learning

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Problem Statement



Detection

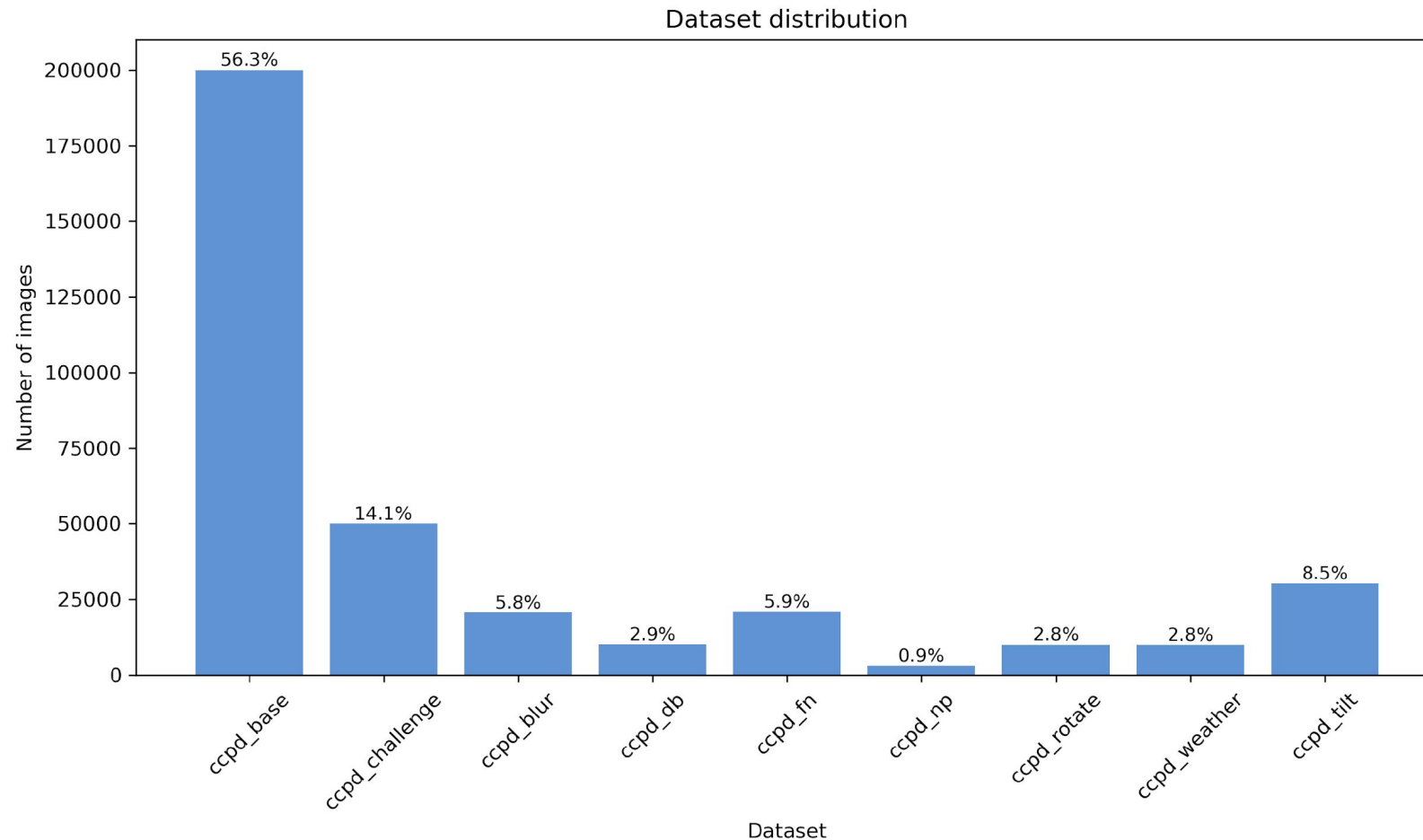


Recognition



皖A WA062

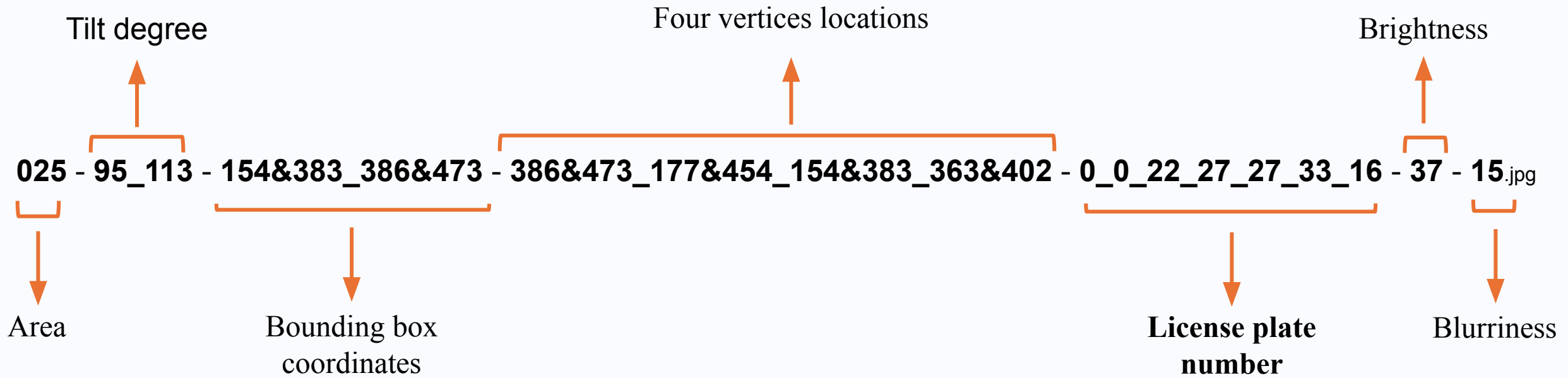
CCPD (Chinese City Parking Dataset)



Total: 300k images

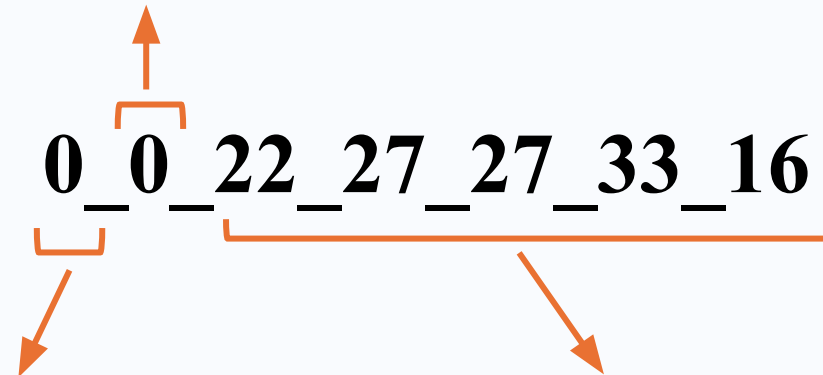
- Different lightning
- Weather conditions
- Different position

Dataset



Dataset

['A', 'B', 'C', 'D', 'E', 'F', 'G', 'H', 'J', 'K', 'L', 'M', 'N', 'P', 'Q', 'R', 'S', 'T', 'U', 'V', 'W', 'X', 'Y', 'Z', 'O']



["皖", "沪", "津", "渝", "冀", "晋", "蒙", "辽",
"吉", "黑", "苏", "浙", "京", "闽", "赣", "鲁",
"豫", "鄂", "湘", "粤", "桂", "琼", "川", "贵",
"云", "藏", "陕", "甘", "青", "宁", "新", "警",
"学", "O"]

['A', 'B', 'C', 'D', 'E', 'F', 'G', 'H', 'J', 'K', 'L', 'M',
'N', 'P', 'Q', 'R', 'S', 'T', 'U', 'V', 'W', 'X', 'Y', 'Z',
'0', '1', '2', '3', '4', '5', '6', '7', '8', '9', 'O']

Baseline Pipeline



Normalized
Bbox

Resize
(224,224)



Detection Model

ResNet34
Feature extractor

Pool2D
adaptive

Regressor
(xmin, ymin,
xmax, ymax)

(64,128)

Recognition Model

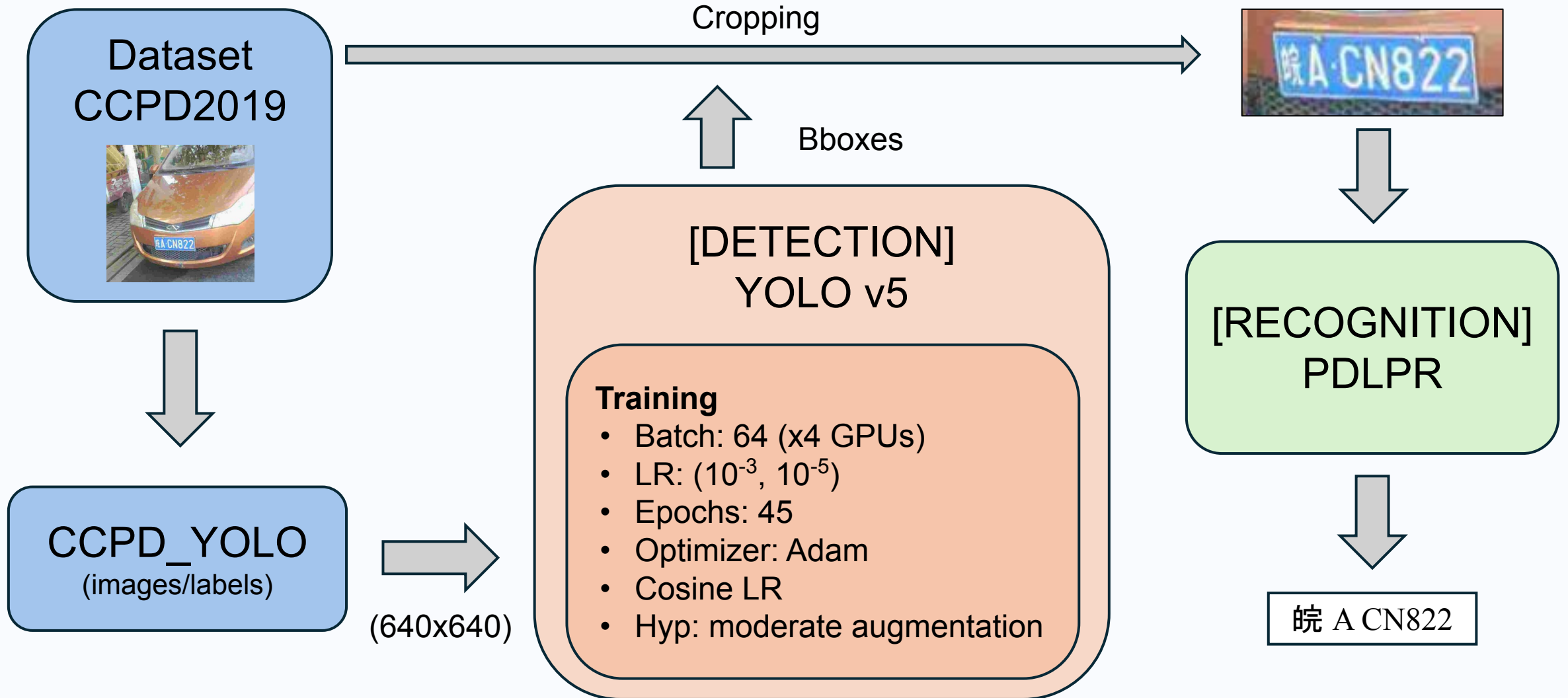
ResNet34
Feature extractor

Classifiers
7 -> one per task

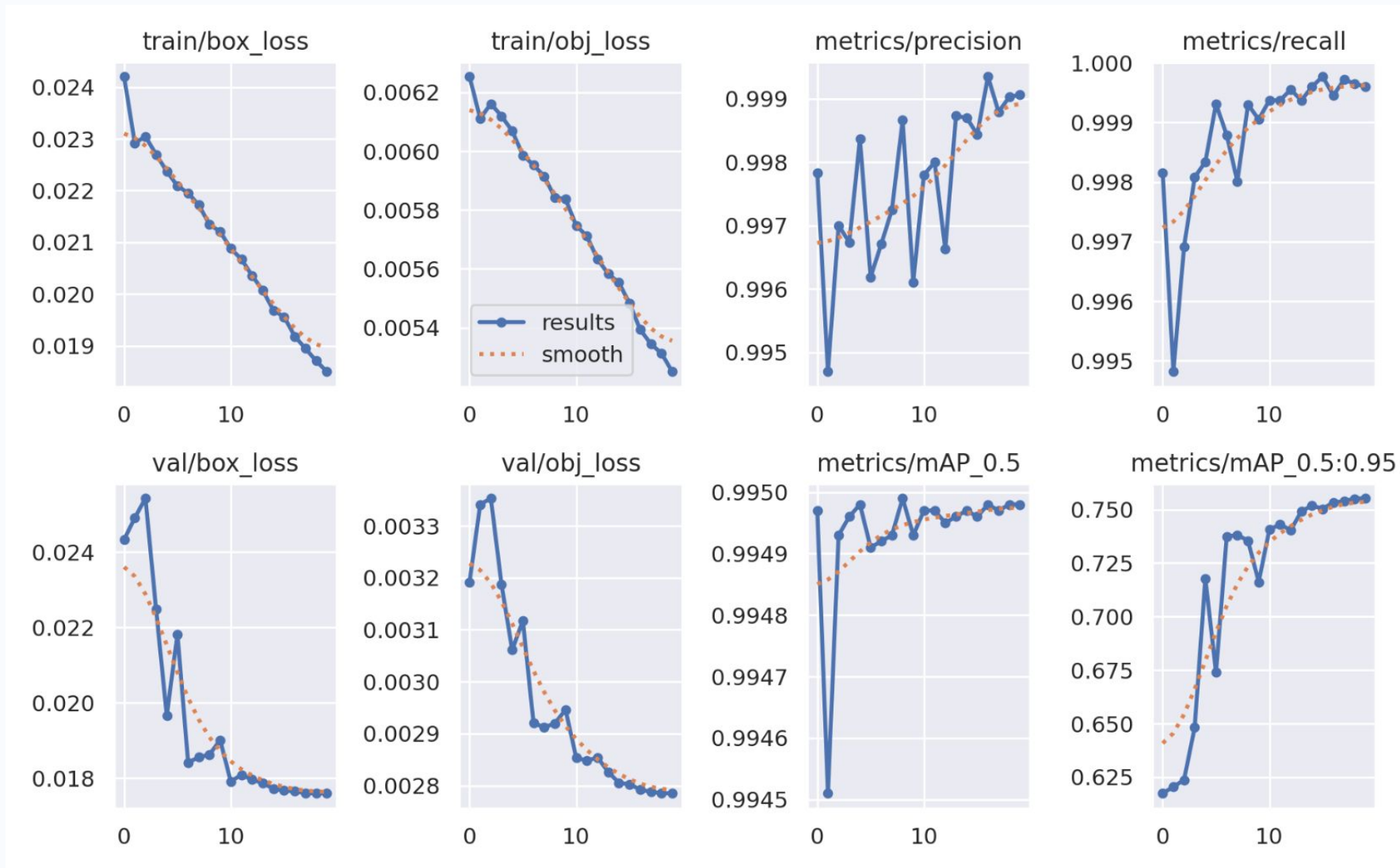
皖A WA062



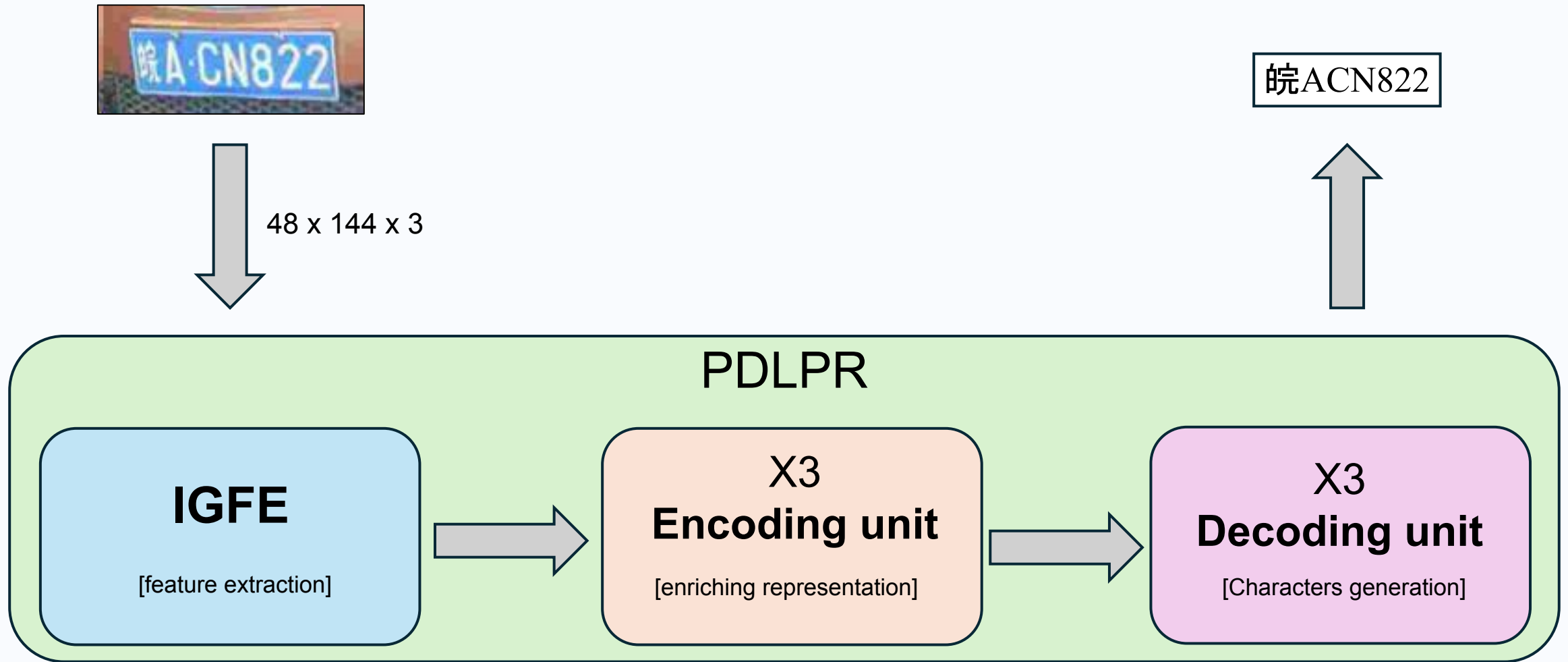
Model Pipeline



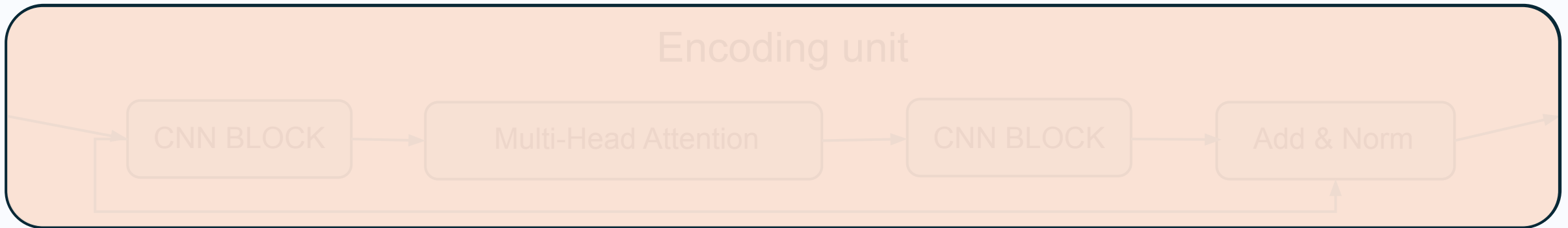
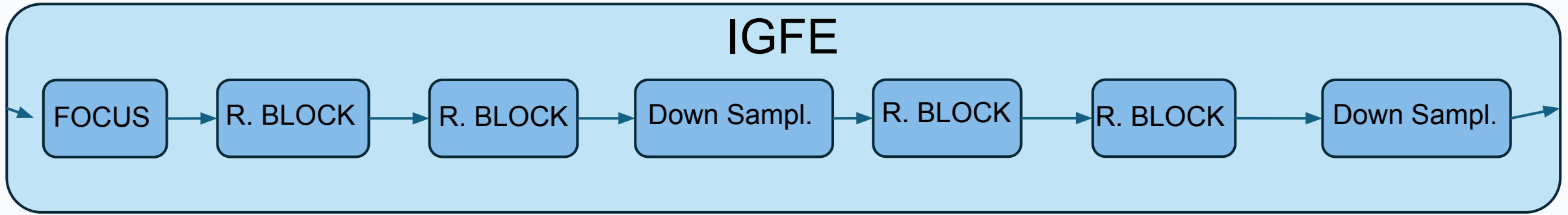
Training [YOLOv5] - last 20 epochs



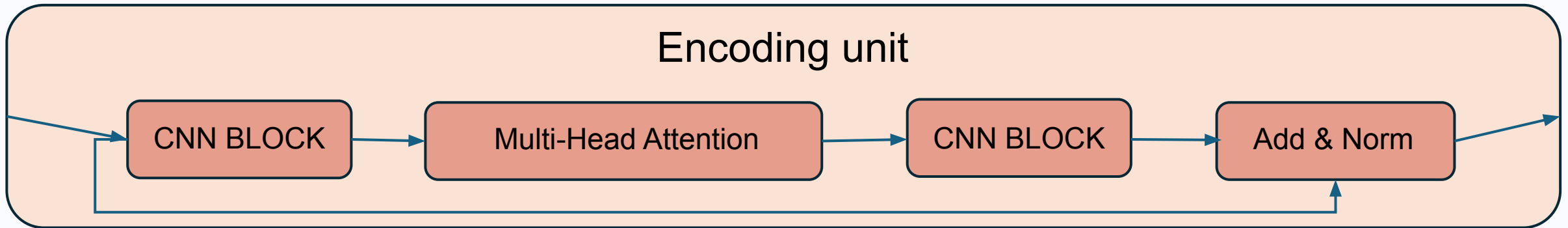
PDLPR [Recognition]



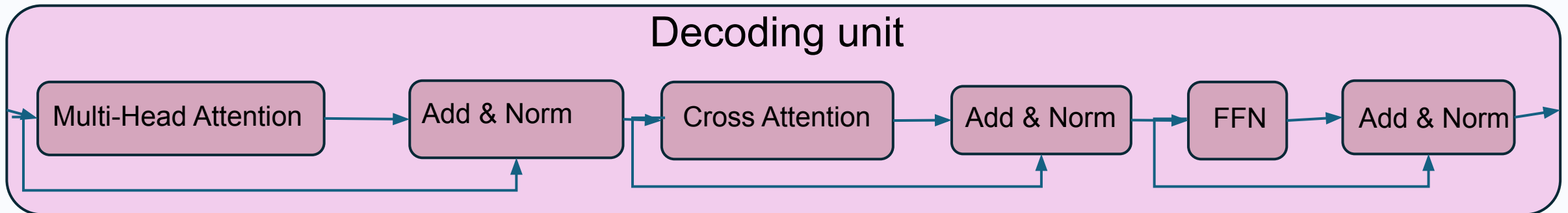
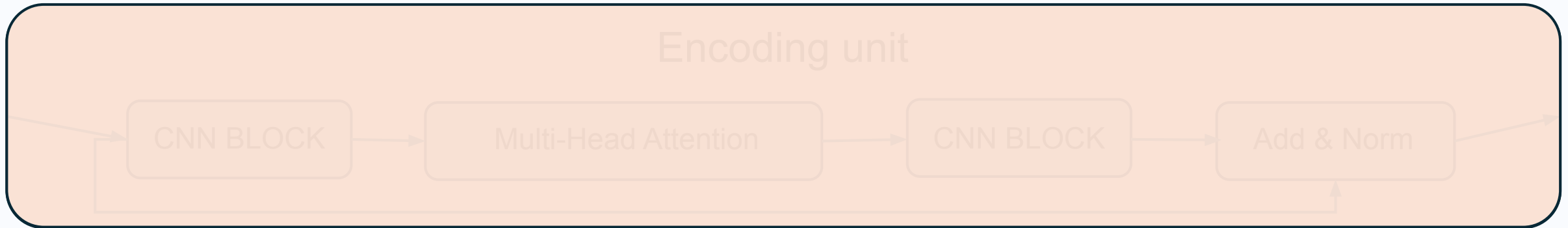
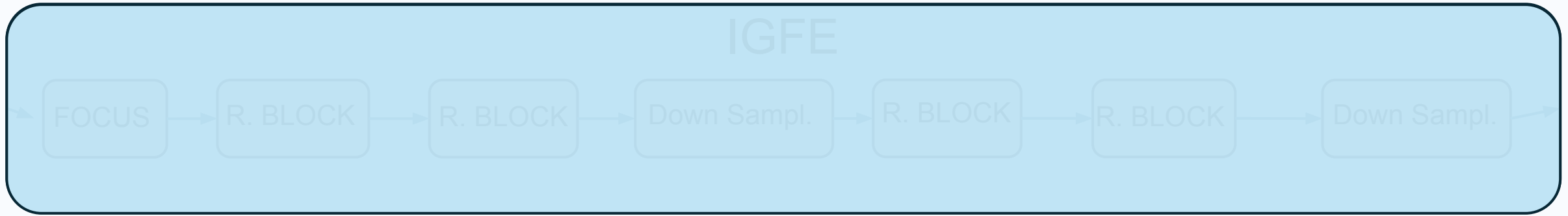
Focus [PDLPR]



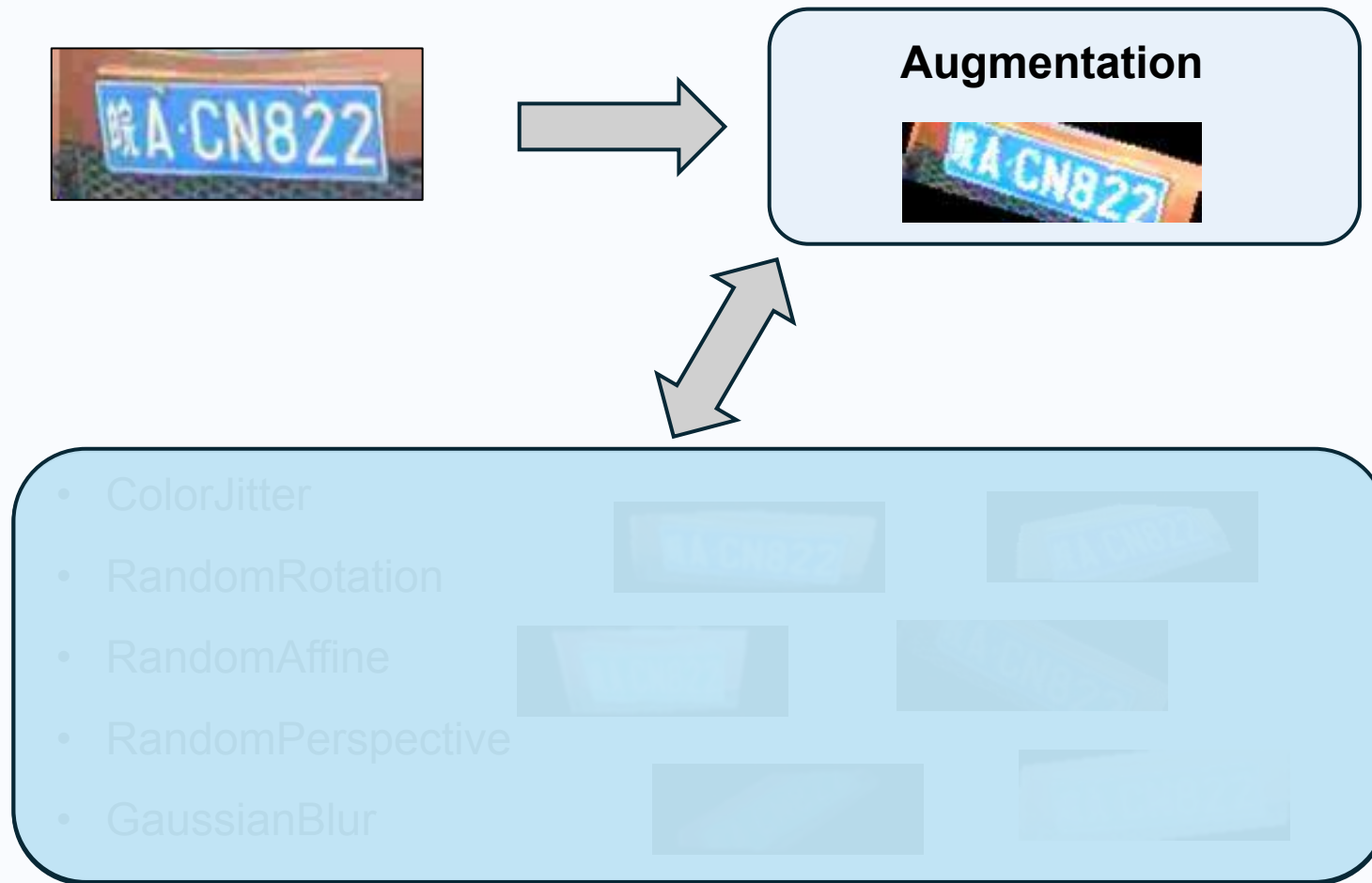
Focus [PDLPR]



Focus [PDLPR]



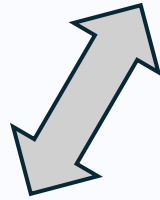
Data Augmentation [PDLPR]



Data Augmentation [PDLPR]



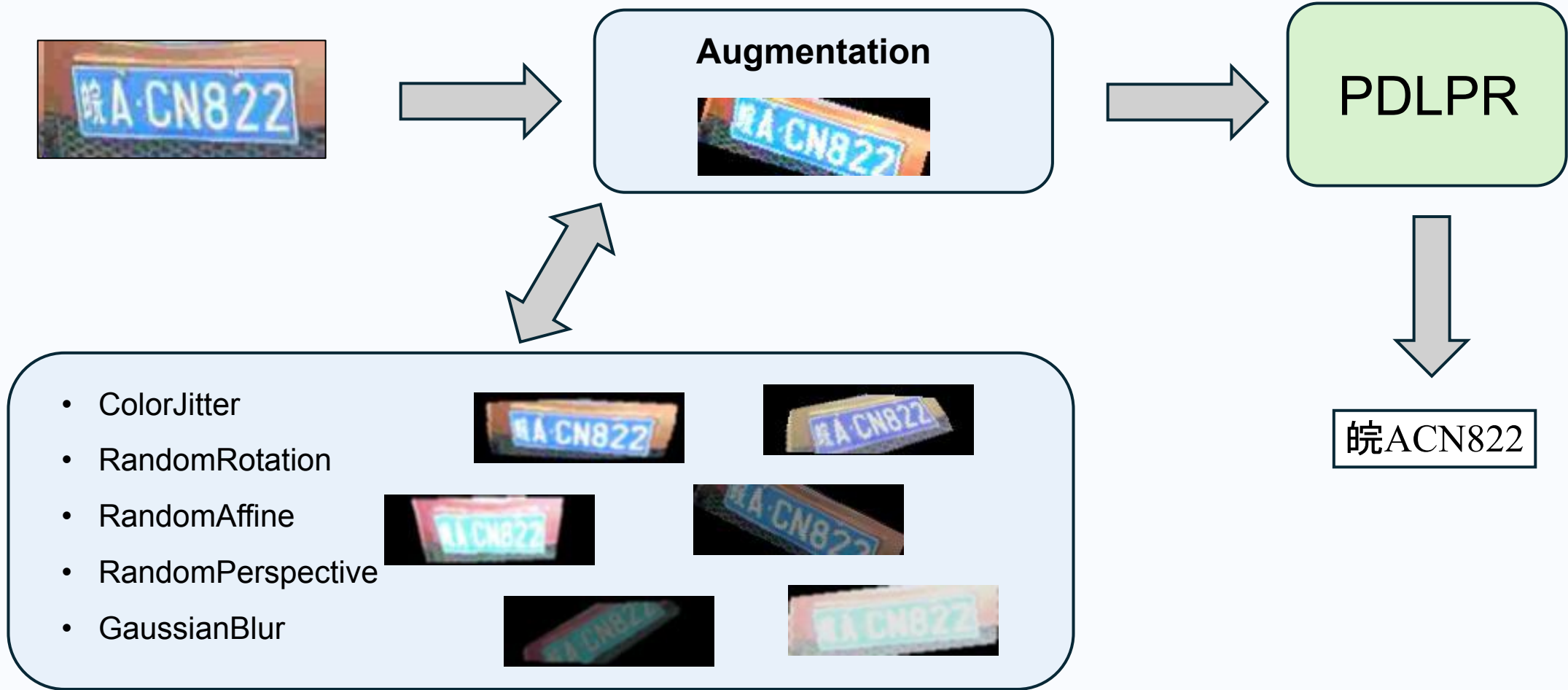
Augmentation



- ColorJitter
- RandomRotation
- RandomAffine
- RandomPerspective
- GaussianBlur



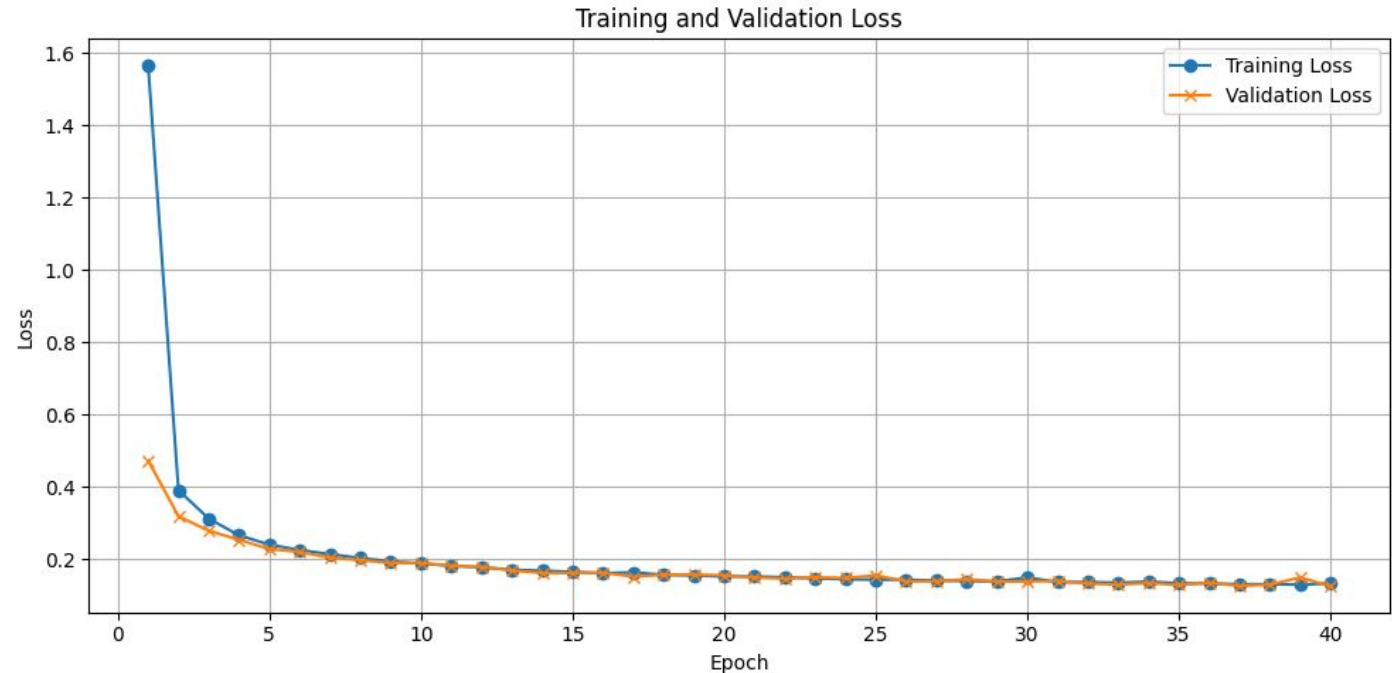
Data Augmentation [PDLPR]



Training and Testing [PDLPR]

Parameters:

- Input dim: $48 \cdot 144 \cdot 3$ [h•w•ch]
- 200k samples (base subset)
- IGFE output: 256 channels
- Encoder emb: 256
- Decoder emb: 256
- Learning rate: 10^{-3}
- Optimizer: Adam
- Epochs: 40



Test Condition	Accuracy (%)
Weather	99.20
Rotate	95.20
Challenge	89.90
FN	89.07
Blur	89.50
Tilt	87.60
DB	86.00

Table 1: Accuracy of PDLPR model trained for 40 epochs on 200K samples

Results

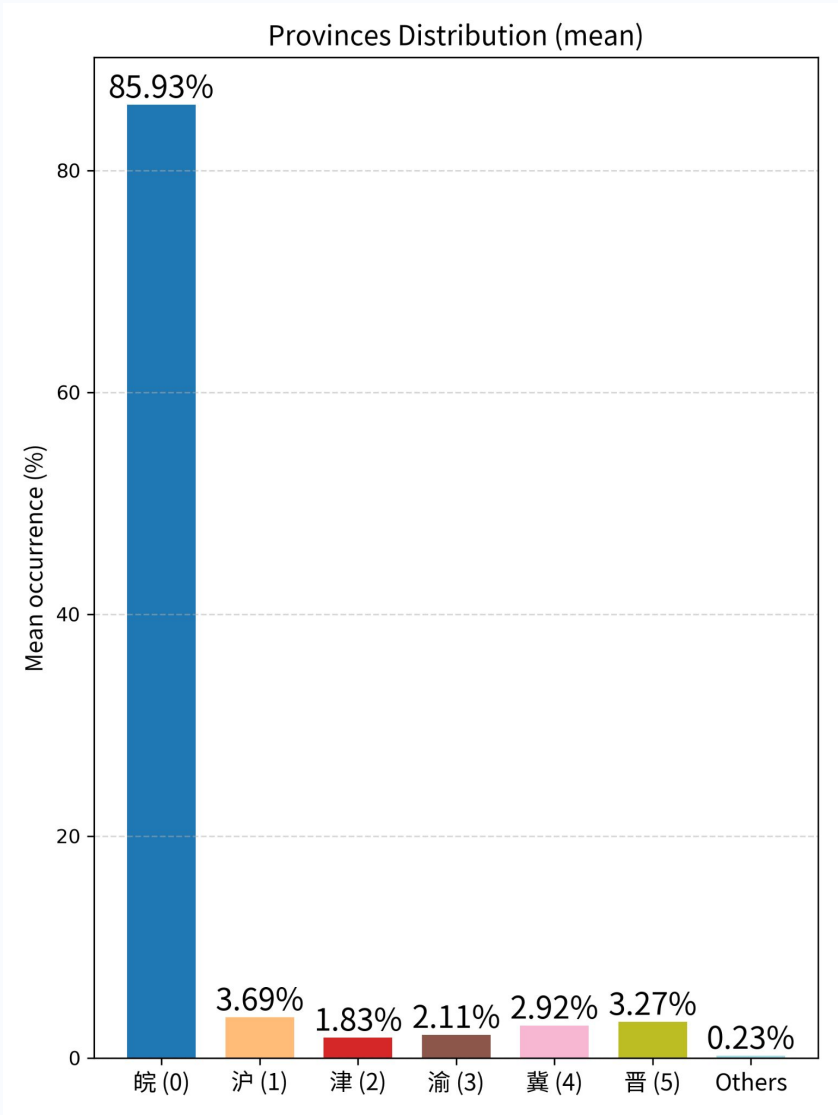
Method	Overall	DB	FN	Rotate	Tilt	Weather	Challenge	FPS
Baseline	70.6	39.6	55.0	91.2	76.7	94.9	66.2	270.3
YOLOv5 (Ours)	83.4	54.9	75.5	95.8	89.5	98.7	86.0	201.6
YOLOv5 (Paper)	96.3	97.7	92.9	98.9	98.9	99.0	90.6	218.3

Table 1: Comparison between the YOLOv5 and Baseline models on the detection.

Method	Overall	DB	FN	Rotate	Tilt	Weather	Challenge	FPS
Baseline	49.2	26.5	40.8	50.1	43.0	95.8	38.9	238.1
YOLOv5-PDLPR (Ours)	81.4	51.7	75.3	94.8	88.1	98.9	79.8	147.6
YOLOv5-PDLPR (Paper)	98.6	99.5	99.5	99.5	99.3	99.4	94.1	159.8

Table 2: Comparison between the YOLOv5-PDLPR and Baseline approaches on the complete pipeline

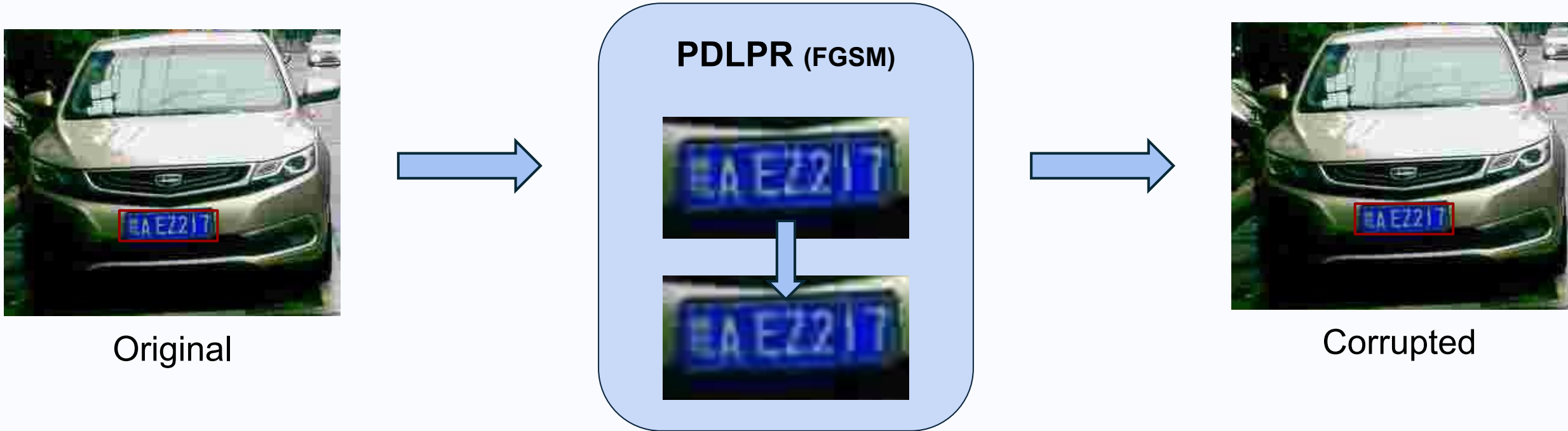
Results



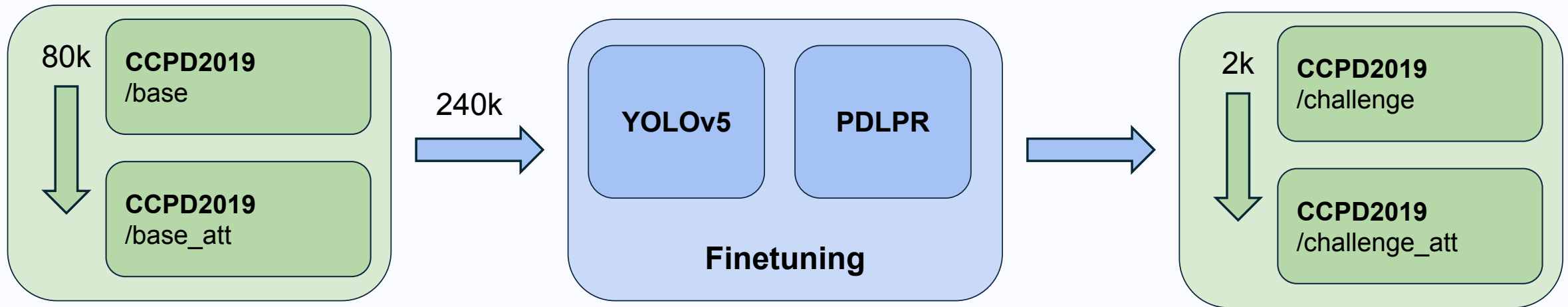
Category	Metric	Baseline	YOLOv5-PDLPR
Overall	Full	49.2	81.4
	W/o-fc	52.8	82.8
	Processed	55.1	91.4
	Processed W/o-fc	59.3	93.0
DB	Full	26.5	51.7
	W/o-fc	29.0	53.7
	Processed	41.4	80.7
	Processed W/o-fc	45.4	83.8
FN	Full	40.8	75.3
	W/o-fc	44.3	76.5
	Processed	50.2	90.4
	Processed W/o-fc	54.7	91.8
Rotate	Full	50.1	94.8
	W/o-fc	55.0	96.0
	Processed	51.7	96.7
	Processed W/o-fc	56.7	97.9
Tilt	Full	42.9	88.0
	W/o-fc	49.3	90.5
	Processed	46.8	91.6
	Processed W/o-fc	53.9	94.0
Weather	Full	95.8	98.9
	W/o-fc	96.3	99.0
	Processed	96.3	99.4
	Processed W/o-fc	96.8	99.5
Challenge	Full	39.0	79.8
	W/o-fc	42.9	81.1
	Processed	44.0	89.4
	Processed W/o-fc	48.6	90.8

Table 4: Accuracy and speed metrics across datasets and perturbation conditions. Comparison between full plate, without first character, and on-process variants.

Novelty: Robustness



Robustness results

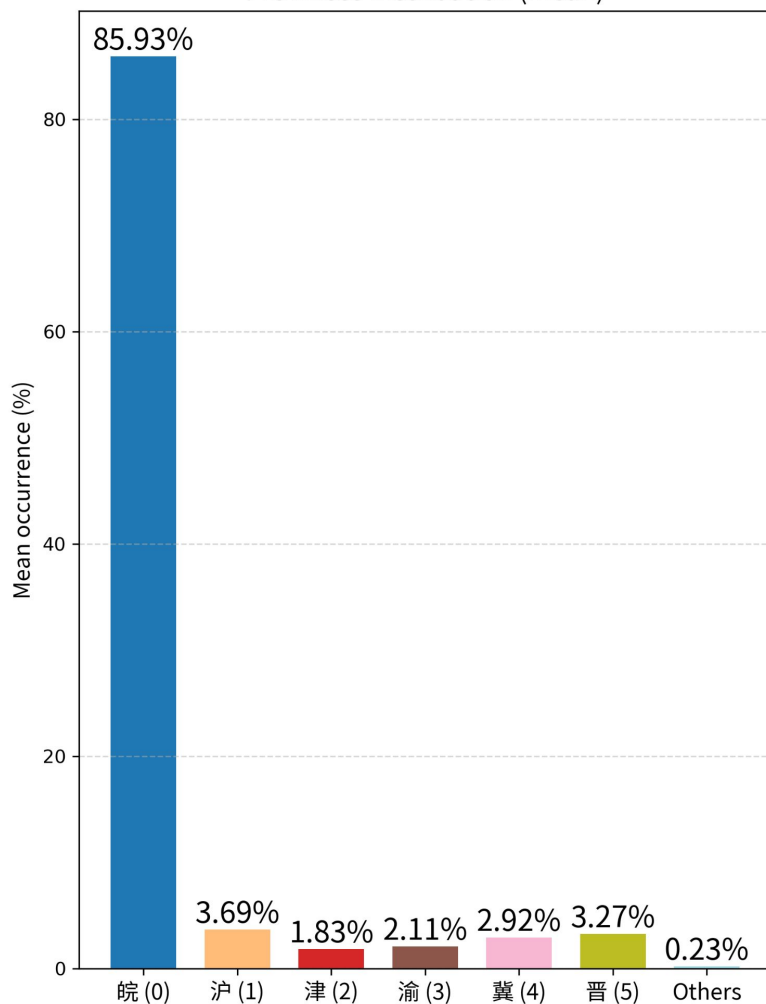


Method	Full ACC	W/o-fc ACC	Proc. Full ACC	Proc. W/o-fc ACC
Base (Clean)	74.20	76.20	83.89	86.15
Robust (Clean)	79.55	81.10	89.94	91.69
Base (Attacked)	40.15	46.85	43.95	51.29
Robust (Attacked)	69.15	71.75	75.70	78.54

Table 3: Comparison of plate recognition accuracy between the base and robust YOLOv5-PDLPR models on clean and attacked data (Challenge testset). Metrics include full plate accuracy, without-first-character accuracy, and their on-process variants.

Future works

Provinces Distribution (mean)



- Balancing the dataset respect to Chinese character with focused data augmentation
- More aggressive data augmentation
- Training not only on base images but also on other subsets
- Training PDLPR with YOLOv5's predictions.

Thanks For Your Attention!

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